

Lingam Hemang

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Professional Summary

Python Engineer who has **designed, owned, and run production data systems in Apple Inc.'s ecosystem for 4 years** – architecting fault-tolerant pipelines on **Spark, Kafka, Airflow, and AWS** that have processed critical data flows without a single data-loss incident. Made independent engineering calls on retry strategy, schema validation, and query optimisation that delivered **40% faster pipelines, 99.5% uptime, and a 35% drop in error rate** – decisions I own end-to-end, not just execute. Now extending that ownership into **GenAI systems** (LangChain, RAG pipelines, AWS Bedrock), shipping agentic tooling from design through deployment.

Education

SRM Institute of Science and Technology

Bachelor of Technology – Electronics and Communication Engineering

May 2018 – June 2022

CGPA: 9.52/10.0

Technical Skills

Languages: Python, SQL, PySpark

Backend & APIs: Django, Flask, REST API Design & Integration, Server-Sent Events (SSE), OOP, SOLID Principles, Design Patterns, pytest

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, Snowflake, AWS S3, AWS Lambda, ETL Pipelines, Semantic Layer & Materialisation, Data Modelling, Schema Design & Validation, Data Quality, Partitioning & Query Optimisation, Tableau

Data Formats: JSON, Parquet, Avro, CSV

GenAI / ML: AWS Bedrock, LangChain, RAG Pipelines, Vector Embeddings, ChromaDB, XGBoost, LightGBM, scikit-learn, Plotly, Dash, LLM, Prompt Engineering

Tools & Platform: Proprietary Spark/Kafka/Airflow/AWS Data Platform, Splunk, Git, Agile/Scrum, Jira, Confluence

Experience

Infosys

Hyderabad, India

Specialist Programmer – Data Engineer

July 2022 – Present

Client: Apple Inc. | Python, Spark, Kafka, Airflow, Snowflake, AWS S3, PySpark, Splunk, Tableau

- Architected production **ETL pipelines** end-to-end on **Apache Spark, Kafka, Airflow, and AWS** – chose **distributed processing and SQL query optimisation** over naive batch loads to ingest internal/external **REST APIs** into **Snowflake**, cutting processing time by **40%**.
- Owned the **semantic layer** from design to production: built materialisation logic in Python, deciding which tables/views to pre-aggregate to balance freshness against **Tableau** query performance – now backs multiple business-critical dashboards.
- Designed and ran **fault-tolerant ingestion pipelines** across **AWS S3** and **Kafka** via **Airflow DAGs**; engineered retry logic and self-healing schema validation from scratch – sustaining **99.5% uptime**, cutting error rate **35%** with zero data-loss incidents in 3 years. Independently productionised a client ML model onto the Airflow cluster.
- Identified a recurring onboarding bottleneck and built **reusable framework extensions** to solve it, cutting onboarding effort **30%**; designed a **Tableau security audit dashboard** from the ground up to surface unauthorised access proactively.
- Owned **data consistency** for a production application – designed the validation logic behind referencing database views and the load process into a **semantic search cluster**; set data governance standards followed across Agile feature teams.

Projects

Agentic AI Git Diff Analyzer | *Python, Django, AWS Bedrock, AWS Lambda, Ollama, ReportLab* | [Repo](#)

- Dual-mode SQL schema impact analyzer** – local (Ollama) and **AWS mode** (Bedrock Claude 3.5 via GitHub Webhook → Lambda → SSM → EC2); commit-accurate SQL scanning with downstream dependency detection across the repo via regex; LLM impact reports with **risk-level badges** (CRITICAL → LOW) as styled A4 PDFs (ReportLab); real-time **Django UI** with **SSE** for live run status and run deduplication by commit hash.

Document Intelligence Chatbot – RAG Pipeline | *Python, Django, LangChain, ChromaDB, Ollama, EasyOCR* | [Repo](#)

- End-to-end **RAG pipeline** (LangChain + ChromaDB + Ollama) with persistent multi-turn conversation memory; multi-format ingestion (PDF, DOCX, TXT, Images) via **PyMuPDF** + **EasyOCR** with auto-rotation; incremental vector store with **timestamped snapshots/rollback**; **10-endpoint Django REST API** supporting live model switching and VectorDB management.

Retail ML Sales Forecasting API | *Python, Flask, scikit-learn* | [Repo](#)

Infosys Hackathon – 2nd Place

- Multi-scenario **ML forecasting system** (Random Forest, XGBoost, LightGBM) for sales, store projections, and product recommendations via **Flask REST API**; cyclical sin/cos month encoding for seasonal patterns; cross-join **purchase probability scoring** across full product catalog; interactive **Dash + Plotly** dashboard with live reactive callbacks.